Extended reading — Week 18: RLHF & PPO for Language Models

May 21, 2026

How to read this list

Each paper below gets: full citation, one-paragraph summary, a *Why it matters for this week* note, and one annotated key passage with brief commentary. The list is ordered chronologically.

1. Schulman et al. 2017 — Proximal Policy Optimization

Citation. J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, *Proximal Policy Optimization Algorithms*, arXiv:1707.06347, 2017.

Summary. Introduces the clipped surrogate $\mathcal{L}^{\text{CLIP}}$ and the adaptive-KL variant $\mathcal{L}^{\text{KLPEN}}$ as first-order, off-the-shelf substitutes for TRPO. Demonstrates competitive performance on continuous control (MuJoCo) and Atari with a tiny implementation footprint. Becomes the default policy-gradient algorithm in deep RL within twelve months of release.

Why it matters for this week. Every line of our PPO implementation is in this paper. The clipped surrogate, the GAE-λ advantages, and the adaptive KL controller all originate here. Even modern RLHF variants like PPOv2 in TRL are conservative re-wrappings of this 2017 algorithm.

Annotated key passage.

> *"By taking the minimum of the clipped and unclipped objective, the final objective is a lower bound (i.e., a pessimistic bound) on the unclipped objective."*

This is the single sentence that justifies Theorem 3.1 ("Why the clip is a pessimistic surrogate") in the notes; it also explains why PPO is robust to large advantages without exploding.

2. Ouyang et al. 2022 — InstructGPT (the foundational RLHF paper)

Citation. L. Ouyang *et al.*, *Training language models to follow instructions with human feedback*, arXiv:2203.02155, NeurIPS 2022.

Summary. The InstructGPT paper. Trains a 1.3B-parameter GPT-3 variant in three stages — SFT on demonstrations, Bradley–Terry reward modelling on human preferences, PPO against the reward model with a KL penalty against the SFT reference — and shows it is preferred by human raters over 175B GPT-3 on a broad range of instruction-following tasks. Forms the template for ChatGPT, Claude, Gemini and essentially every subsequent production aligned LLM.

Why it matters for this week. The three-stage pipeline (the three-stage pipeline in `notes.pdf`, Algorithm 1) is theirs. Their Figure 5 (preference win-rate vs. model size) is the empirical justification for the entire field. They are also the first to publicly enumerate the specific failure modes — sycophancy, refusal collapse, length bias — that we'll spend weeks 18–22 mitigating.

Annotated key passage.

> *"We find that the outputs of our 1.3B InstructGPT model are preferred to those of the 175B GPT-3, despite having 100x fewer parameters."*

The headline result. The implication — that an alignment-trained small model can dominate a much larger base model on instruction-following — shifts the whole research economics of language modelling toward post-training.

3. Gao, Schulman & Hilton 2023 — Scaling laws for reward overoptimization

Citation. L. Gao, J. Schulman, J. Hilton, *Scaling Laws for Reward Model Overoptimization*, PMLR v202 (ICML 2023).

Summary. Constructs a controlled environment in which a "gold" reward model is treated as ground truth and a smaller "proxy" RM is fit on samples from it. PPO is then run against the proxy while gold reward is measured on held-out data. Finds an analytic functional form: $R_{\text{gold}}(d) \approx d(\alpha - \beta d)$ for KL distance $d = \sqrt{\mathrm{KL}(\pi_\theta \| \pi_{\text{ref}})}$, with both α and the optimal $d^\star$ scaling smoothly with proxy-RM size and dataset size.

Why it matters for this week. This is *the* paper on reward hacking. It quantifies precisely why a KL leash and early stopping are not optional. The schematic figure in the notes is a stylised version of their Figure 1.

Annotated key passage.

> *"We find that the gold reward, $R(d)$, can be very accurately fit by a function of the form $R(d) = d(\alpha_{bon} - \beta_{bon} d)$ for best-of-n sampling, and $R(d) = d(\alpha_{RL} - \beta_{RL} \log d)$ for reinforcement learning."*

The two functional forms differ — RL is logarithmic in d, BoN linear — and the RL form decays *faster* once KL is past its sweet spot. This explains why brute-force PPO without gold-eval is dangerous in production: you cannot see the gold drop without independent evaluations.

4. Bai et al. 2022 — Constitutional AI and RLAIF

Citation. Y. Bai *et al.* (Anthropic), *Constitutional AI: Harmlessness from AI Feedback*, arXiv:2212.08073, 2022.

Summary. Replaces the human-preference half of RLHF with *AI feedback*. Two stages: (i) the model critiques and revises its own outputs against a written "constitution"; (ii) a separate preference model is trained on AI-generated comparisons and used to drive the RL loop, exactly as a human-RM would. The result, Claude's harmlessness profile, is the first credible evidence that scalable oversight via AI feedback can replace human feedback for non-edge-case alignment.

Why it matters for this week. Same machinery as InstructGPT — PPO, KL leash, BT loss — but a completely different source of supervision. Sets the stage for week 19's iterative DPO and week 20's weak-to-strong supervision.

Annotated key passage.

> *"[CAI] reduces the reliance on human supervision while preserving the ability to fine-tune for harmlessness, suggesting a path to scalable oversight of AI systems."*

The plausibility claim of the entire scalable-oversight agenda. The 2024–2026 literature (process reward models, deliberative alignment, debate, etc.) all live downstream of this paper.

5. Lambert et al. 2024 — Tülu 3

Citation. N. Lambert *et al.*, *Tülu 3: Pushing Frontiers in Open Language Model Post-Training*, arXiv:2411.15124, 2024.

Summary. The first fully open RLHF+RLVR recipe at production scale: a complete SFT → DPO → RLVR (PPO with verifiable rewards) pipeline for Llama-3.1, with open datasets, open code, and an open evaluation suite. Closes the gap with proprietary models on a wide range of benchmarks while making every artefact reproducible.

Why it matters for this week. The closest thing in 2026 to a "how to run RLHF on a real model, from scratch, end-to-end" reference. The dataset (`tulu-3-sft-mixture`, `tulu-3-pref-mixture`) is what we'd reach for on a serious RLHF experiment. The TRL recipe in `project/trl_recipe.py` is a scaled-down version of theirs.

Annotated key passage.

> *"We provide a recipe that achieves competitive performance with frontier closed models such as GPT-4o-mini and Claude 3.5 Haiku across a wide range of benchmarks, while fully releasing our data, model weights, training and evaluation code."*

Open-source RLHF caught up with closed-source RLHF in 2024. The lesson is that the algorithmic core has stabilised — the remaining differentiator is data quality, not method.

6. Christiano et al. 2017 — Deep RL from Human Preferences

Citation. P. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, D. Amodei, *Deep Reinforcement Learning from Human Preferences*, arXiv:1706.03741, NeurIPS 2017.

Summary. The original RLHF paper. Trains an RL agent on Atari and MuJoCo tasks using only pairwise human preferences over trajectory segments — no hand-engineered reward function. Lays out the three-stage pipeline (preference collection, RM training, RL against the RM) that would become the InstructGPT template five years later.

Why it matters for this week. The pipeline architecture we use today was invented *here*, in 2017, on Atari, by the same OpenAI–DeepMind team that would later build ChatGPT. Reading this paper alongside InstructGPT is the cleanest way to see the continuity of ideas across the deep-RL-to-LLM transition.

Annotated key passage.

> *"We show that this approach can effectively solve complex RL tasks without access to the reward function, using less than 1% of the feedback that the agent receives."*

The astonishing data efficiency point: a few hundred human comparisons could replace millions of environment-reward samples. The same calculus is what makes preference RL economical for LLMs.

7. Ziegler et al. 2019 — Fine-tuning language models from human preferences

Citation. D. M. Ziegler, N. Stiennon, J. Wu, T. B. Brown, A. Radford, D. Amodei, P. Christiano, G. Irving, *Fine-Tuning Language Models from Human Preferences*, arXiv:1909.08593, 2019.

Summary. The first paper to apply RLHF to a language model (GPT-2). Pioneered the practical machinery — KL-shaped per-token rewards, value-head sharing the policy trunk, adaptive KL — that would later go into InstructGPT.

Why it matters for this week. This is the paper that turned RLHF from an RL idea (Christiano 2017) into an LLM methodology. Almost every implementation detail you'll meet in TRL or OpenRLHF was invented here.

Annotated key passage.

> *"We use a KL constraint to keep the policy from drifting too far from the supervised initialisation; without it, the policy quickly collapses to a degenerate solution that the reward model gives high score to but that humans rate as poor."*

A clear early statement of reward-model overoptimisation, written before Gao et al. 2023 quantified it.

8. Stiennon et al. 2020 — Learning to summarize from human feedback

Citation. N. Stiennon, L. Ouyang, J. Wu, D. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, P. Christiano, *Learning to Summarize from Human Feedback*, arXiv:2009.01325, NeurIPS 2020.

Summary. The TL;DR-summarisation benchmark for RLHF. Trains a 6.7B model with PPO against a learned reward model on human comparisons of news-article summaries. The RLHF-trained model is preferred to a 100× larger supervised baseline.

Why it matters for this week. The bridge between Ziegler 2019 (proof of concept) and Ouyang 2022 (industrial scale). Also the source of the much-cited "reward-model overoptimisation" plot, where held-out human ratings degrade as RM-score keeps rising.

Annotated key passage.

> *"As training progresses, the policy's reward according to the reward model continues to improve, but its quality according to human labels saturates and eventually degrades. This is a clear indication of reward-model overoptimisation and motivates the KL constraint in our algorithm."*

The smoking-gun figure for the entire reward-hacking literature. Read this paper carefully before deciding how to length-regulate your own runs.

Pointers for further depth

- Schulman et al. 2016 — *Generalized Advantage Estimation*, arXiv:1506.02438. The GAE bias-variance dial; if you want to understand the λ knob in your RLHF runs, read this.
- Casper et al. 2023 — *Open Problems and Fundamental Limitations of RLHF*, arXiv:2307.15217. A clear-eyed survey of why RLHF is much harder than it looks.
- Coste et al. 2023 — *Reward Model Ensembles Help Mitigate Overoptimization*, arXiv:2310.02743. Ensembles of RMs vote; the policy is rewarded only when they agree.
- Razin et al. 2024 — *Likelihood Displacement*, arXiv:2410.08847. The technical explanation of why DPO and PPO both decrease $\log \pi(y_l)$ aggressively.
- Lambert — *The RLHF Book* (online, 2026). The best free reference text on the entire pipeline, regularly updated.

Sourced from

`research/llm_rl_sota.md` §1 (Classical RLHF), §6 (Constitutional AI / RLAIF), §7 (reward hacking, KL control), and §11 (key reading list).